

# ATEEB AHMED KHAN

Machine Learning Engineer · Computer Vision & LLM/RAG Systems  
ateebkhan669@gmail.com · Maharashtra, India · [LinkedIn](#) · [GitHub](#) · [Portfolio](#)

ML Engineer building and deploying Computer Vision and LLM/RAG applications end-to-end, from data preparation and model training to production inference. Built a YOLO11 MRI tumor detector (95.1% mAP@0.5), a DenseNet201 face mask classifier (98.99% accuracy), and a RAG system using LangChain, ChromaDB, and openai/gpt-oss-120b. Proficient in Python, TensorFlow, and PyTorch.

## EXPERIENCE

**AI/ML Mentor** — DSIAR Tech 07/2025 – Present · Remote, Part-Time

- Built a 30-module, 120–140 lesson production-oriented AI/ML curriculum with capstone deployment projects; a Deep Learning track is in active development.
- Translated model evaluation, feature engineering, and inference optimization theory into modular, project-driven teaching units.

**AI/ML Engineer (Intern)** — Labmentix 07/2025 – 12/2025 · Remote

- Developed and deployed end-to-end ML pipelines for data ingestion, preprocessing, model training, and production inference using modular, object-oriented Python components designed for reproducibility and reuse.
- Deployed ML models as interactive web applications on Google Colab + Streamlit, integrating TensorFlow inference with a functional UI layer and prediction logging to track output behavior post-deployment.
- Designed reproducible training pipelines with automated validation checkpoints, improving model reliability across iterative builds and eliminating manual evaluation steps.

**Machine Learning Intern** — Feynn Labs 09/2022 – 11/2022 · Remote

- Developed ML-driven market segmentation models for EV industry analysis, applying feature engineering and clustering to structure raw trend data into actionable product prototype inputs, informing two AI product scoping decisions.
- Built and evaluated end-to-end preprocessing and training pipelines in Python using Scikit-learn and Pandas.

## PROJECTS

**Document Q&A RAG System** [Live Demo ↗](#)

LangChain · ChromaDB · openai/gpt-oss-120b · Groq API · all-MiniLM-L6-v2 · PyMuPDF

- Built and deployed an end-to-end RAG pipeline from PDF ingestion and chunking to semantic retrieval and LLM generation, deployed as a publicly accessible Streamlit application.
- Engineered a modular 4-component system (ingest, retriever, generator, UI) enabling independent testing and redeployment of each stage; reduced end-to-end query response time via async retrieval and Groq's inference API.
- Selected all-MiniLM-L6-v2 over larger embedding models after benchmarking retrieval precision vs. latency trade-offs on domain-specific document corpora.

**Brain Tumor Detection — YOLO11** [Live Demo ↗](#)

YOLO11 · PyTorch · Streamlit · Medical Imaging

- Designed and deployed a real-time MRI tumor detection system achieving 95.1% mAP@0.5 and 92.7% precision, with sensitivity-tuned thresholds to minimize false negatives in a clinical-risk context, prioritizing recall over precision by design.
- Built an end-to-end inference pipeline from MRI scan input through YOLO11 object detection to annotated output rendering, deployed as a low-latency web application with sub-second inference per scan.

**Real-Time Face Mask Detection** [Live Demo ↗](#)

DenseNet201 · TensorFlow Lite · Mediapipe · Streamlit

- Trained a DenseNet201 classifier on a 992-image dataset achieving 98.99% accuracy; compressed and converted to TensorFlow Lite for edge-optimized inference, reducing model size for resource-constrained environments.
- Integrated Mediapipe face detection as a preprocessing stage into a unified inference pipeline, enabling real-time video stream classification at 30fps with a modular, production-ready architecture.

## EDUCATION

**MSc Computer Science** — IU International University of Applied Sciences, Berlin 2023 – /2026

Distance Learning · GPA: 1.8 (German Scale) · Thesis: “Brain Tumor Detection Using Vision Transformers and XAI”

**B.Tech Robotics & Automation** — MIT World Peace University, Pune 2018 – 2022

GPA: 8.89 / 10

## SKILLS

**Core Engineering:** Python · SQL · OOP · Docker · Git · Flask · GitHub

**ML / Deep Learning:** TensorFlow · PyTorch · Scikit-learn · Keras · Feature Engineering · NumPy · Pandas

**Generative AI & LLMs:** LangChain · RAG · ChromaDB · LLM · Prompt Engineering · Embedding Models

**Computer Vision & Data:** OpenCV · MediaPipe · YOLO · ViT (Vision Transformers)

**Deployment & Applications:** Streamlit · TFLite

## CERTIFICATIONS

Machine Learning Specialization (Coursera / Stanford Online) · Google Data Analytics (Coursera / Google) · Deep Learning & OpenCV for Computer Vision (Udemy)